

```
!pip install transformers torch
```

```
from transformers import pipeline
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (4.46.2)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.5.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (3.16.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (24.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (2024.11.6)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (0.20.1)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (0.5.1)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (4.67.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (4.12.2)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (75.1.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (3.4.1)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (3.1.4)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (2024.12.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (0.27.2)
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (2.0.1)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (4.6.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (2024.12.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (1.0.6)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (3.10)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (8.1.8)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (0.1.2)
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
```

```
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")
```

```
# Addressing the warning: "Please make sure the generation config includes `forced_
# This sets the beginning-of-sequence token ID to be forced as the first generated
model.generation_config.forced_bos_token_id = tokenizer.bos_token_id
```

```
Loading weights: 100% 511/511 [00:00<00:00, 644.62it/s, Materializing param=model.encoder.layers.11.se
```

```
text = """Artificial Intelligence (AI) is transforming industries by enabling machines to learn and make decisions. It is widely used in healthcare, finance, transportation, and education. AI helps improve efficiency and reduces human effort. Many companies use AI for automation and better decision-making. However, there are challenges such as ethical concerns, privacy issues, and job displacement. Researchers are working to make AI systems more transparent, explainable, and trustworthy, so that they can benefit society without causing harm."""
```

```
inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")

# Short summary
short_summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=30,
short_summary_text = tokenizer.decode(short_summary_ids[0], skip_special_tokens=True)

# Medium summary
medium_summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=50,
medium_summary_text = tokenizer.decode(medium_summary_ids[0], skip_special_tokens=True)

# Long summary
long_summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=80,
long_summary_text = tokenizer.decode(long_summary_ids[0], skip_special_tokens=True,

print("Short Summary:", short_summary_text)
print("\nMedium Summary:", medium_summary_text)
print("\nLong Summary:", long_summary_text)
```

```
Short Summary: Artificial Intelligence (AI) is transforming industries by enabling machines to learn and make decisions. It is widely used in healthcare, finance, transportation, and education. AI helps improve efficiency and reduces human effort. Many companies use AI for automation and better decision-making. However, there are challenges such as ethical concerns, privacy issues, and job displacement. Researchers are working to make AI systems more transparent, explainable, and trustworthy, so that they can benefit society without causing harm.

Medium Summary: Artificial Intelligence (AI) is transforming industries by enabling machines to learn and make decisions. It is widely used in healthcare, finance, transportation, and education. AI helps improve efficiency and reduces human effort. Many companies use AI for automation and better decision-making. However, there are challenges such as ethical concerns, privacy issues, and job displacement. Researchers are working to make AI systems more transparent, explainable, and trustworthy, so that they can benefit society without causing harm.

Long Summary: Artificial Intelligence (AI) is transforming industries by enabling machines to learn and make decisions. It is widely used in healthcare, finance, transportation, and education. AI helps improve efficiency and reduces human effort. Many companies use AI for automation and better decision-making. However, there are challenges such as ethical concerns, privacy issues, and job displacement. Researchers are working to make AI systems more transparent, explainable, and trustworthy, so that they can benefit society without causing harm.
```

```
texts = [
    """Artificial Intelligence is widely used in healthcare for diagnosis and treatment. It helps doctors analyze medical data quickly and accurately.""",
    """India has made significant achievements in space exploration through ISRO missions, showcasing cost-effective innovation.""",
    """Once upon a time, a young girl dreamed of becoming a scientist. She worked hard and finally achieved success through determination.""",
    """Microservices architecture allows developers to break applications into small, manageable pieces, improving scalability but increasing system complexity.""",
    """Hey, are we meeting tomorrow to complete the project? We should finalize everything by then.""",
]
```

```
texts = [text] # Define texts as a list containing the existing 'text' variable

for i, t in enumerate(texts):
    inputs = tokenizer(t, max_length=1024, truncation=True, return_tensors="pt")
    # Using model.generate and tokenizer.decode instead of summarizer
```

```
summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=40, min_length=10)
summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

print(f"\n--- Text {i+1} ---")
print("Original:", t)
print("Summary:", summary_text)
```

--- Text 1 ---

Original: Artificial Intelligence (AI) is transforming industries by enabling machines to learn from data and make decisions. It is widely used in healthcare, finance, transportation, and education. AI helps improve efficiency and reduces human effort. Many companies use AI for automation and better decision-making. However, there are challenges such as ethical concerns, privacy issues, and job displacement. Researchers are working to make AI systems more transparent, fair, and accountable so that they can benefit society without causing harm.

Summary: Artificial Intelligence (AI) is transforming industries by enabling machine

long_text = """Artificial Intelligence (AI) has become one of the most important technologies in the world. It is used in various industries such as healthcare, finance, education, and transportation. AI helps process large amounts of data and make decisions faster than humans. In healthcare, AI helps doctors recommend treatments. In finance, it is used for fraud detection and risk management. AI is also used in self-driving cars and smart cities.

Despite its advantages, AI has several challenges. Ethical concerns such as bias in algorithms and privacy issues are major issues. Many people are worried about job loss due to automation. Researchers are working to make AI systems more transparent, fair, and accountable.

AI continues to evolve rapidly, and its future depends on how well we address these challenges. We need to ensure that AI is used responsibly and that its benefits are shared by all of society.

```
inputs = tokenizer(long_text, max_length=1024, truncation=True, return_tensors="pt")
long_summary_ids = model.generate(inputs["input_ids"], num_beams=4, max_length=120, min_length=10)
long_summary_text = tokenizer.decode(long_summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

print("Long Text Summary:", long_summary_text)
```

Long Text Summary: Artificial Intelligence (AI) has become one of the most important

